Eigenvalues

Exercise 1: Determine the eigenvalues of matrix

$$A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Solution:

Eigenvalues

Exercise 2: Find the sum and product of Eigenvalues

$$A = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & -6 \end{bmatrix}$$

Solution:

Eigenvalues

Exercise 3: Given product of two Eigenvalue of the matrix is 16. Find the third value of Eigenvalue of A

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Solution:

Eigenvalues

Exercise 4: Two eigenvalues of matrix are 3 and 0, what is the third eigenvalue? Show product of eigenvalues = 0

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Solution:

Polynomial Method

Exercise: Determine the eigenvalues and eigenvectors for following matrix

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Solution:

Polynomial Method

Exercise 2: Determine the eigenvalues and eigenvectors for following matrix

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

Solution:

Power Method

Exercise: Determine dominant eigenvalue and its eigenvector for matrix

$$A = \begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & -3 \\ 0 & -1 & 1 \end{bmatrix}$$

With $x^0 = [1,1,1]^T$ using power method, in 4 decimal and error of 0.1

Solution:

Shifted Power Method

Exercise: Find the smallest absolute eigenvalue

$$A = \begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & -3 \\ 0 & -1 & 1 \end{bmatrix}$$

With $x^0 = [1,1,1]^T$ using power method, in 4 decimal and error of 0.1

Solution: